SAMPLE

Famous Astronomers: Margaret Burbidge

Stellar Nucleosynthesis Pioneer

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Introduction

This volume belongs to the Famous Astronomers shelf, the biographies of the men and women who measured the sky. The subject here being Margaret Burbidge, the British-born spectroscopist who helped explain where the chemical elements come from, and the woman whose career doubled as the record of the professions' long argument about itself. Readers arriving from the companion lives will recognize the room's arrangement; anyone starting cold loses nothing by beginning at the first chapter and walking straight through.

The book's method being the shelf's custom: a single unbroken account, no footnote machinery, no lifted almanac text, sums left in plain sight and stories kept within the record's own bounds. The spine of dates holds the whole account up: the birth in Cheshire in August nineteen nineteen, the London schooling, the University College degrees of nineteen thirty-nine and nineteen forty-three, the wartime caretaking at the University of London Observatory, the marriage of nineteen forty-eight, the immigration of nineteen fifty-five, the synthesis paper of nineteen fifty-seven, the galaxy spectroscopy of the sixties, the royal observatory directorship of nineteen seventy-two, the society presidency of the late seventies, the San Diego decades, and the death at one hundred in April twenty twenty. Every chapter hangs its incidents on those dates, which a curious reader may check against the sources named in the back of the book.

The woman the dates describe being a singular composite: the child who worked out the timing of her own conception against the Armistice, the student who kept the observatory running through the blackout, the researcher denied the mountain because of her sex and later flown to it, the analyst who co-authored the century's most-cited paper on the making of matter, the director who walked out of the field's oldest institution rather than preside over what she judged a mistake, and the president who told her own profession which states it would not meet in. The volume holding nothing that the record cannot carry, and carrying the whole record.

The chapters following the library's standard tour, from discovery through understanding and characteristics, through the science and the observation, the features, the context, the modern assessment, the research, the discoveries, the challenges, the future, the trivia, and the significance. One thread running through all fourteen: the light having been the instrument, the woman having read it better than almost anyone alive, and the reading having changed what the species knows it is made of.

The subject having been chosen for this shelf on the standard grounds and one unusual one: the standard grounds being the synthesis paper's permanence and the century's length, the unusual one being the career's double duty as the era's own case study, and the biography being the shelf's most complete intersection of the science and the sociology. The volume being written for the reader who wants both accounts in one telling, the wanting being legitimate and the supplying being this book's own assignment, the assignment being carried out in the chapters that follow and the following being recommended in order though the order being the reader's own affair.

A last word on the sources: the life having been unusually well documented by the subject's own cooperation, the interviews and the memoirs having been given freely in the later decades, and the volume being able to quote the woman on every phase of the woman's own century, the quotation being the biographer's luxury and the luxury having been earned by the subject's candor, the candor being the record's own hallmark and the hallmark showing on every page, the page being the reader's and the reader being welcomed as the volume's last collaborator, the collaboration being the standard offer of the shelf and the offer standing throughout the chapters that follow.

Chapter 1: The Discovery of Margaret Burbidge

The discovery begins with the birth and its arithmetic. Eleanor Margaret Peachey having been born on the twelfth of August, nineteen nineteen, near Davenport in Cheshire, the daughter of Stanley John Peachey and Margaret Scott Peachey, and the child having kept a private joke about her own origin all her life: the birth falling nine months after the Armistice of November nineteen eighteen, and the girl having concluded, with the precision she later spent on stellar spectra, that she had probably been conceived amid the celebration of the war's end. The anecdote being the first exhibit of the volume because the anecdote being the subject in miniature, the mind that would count the abundances of the elements having begun by counting the months to her own beginning.

The family having moved to London in nineteen twenty-one, the father's patent connected to the vulcanization of rubber having paid for the move and for the industrial chemistry laboratory he set up there, and the household being the child's first laboratory. The mother having been housekeeper before her marriage, both parents being devout Roman Catholics of modest schooling, and the home having valued work and precision without valuing

pedigree, the combination suiting a future scientist better than the reverse.

The stars having entered through a grandfather's gift or a London fog or both, the childhood accounts varying, the fixed point being the total solar eclipse of June nineteen twenty-seven, visible in Britain when the girl was seven, and the spectacle having set the compass. The mathematics having come easily at the schools she attended, the support of her mother being remembered as crucial, and the University College London entrance of nineteen thirty-six arriving when the girl was sixteen and already certain of the destination.

The discovery of the woman being partly the discovery of the war's timing. The undergraduate degree having arrived in nineteen thirty-nine, the doctorate in nineteen forty-three, both from University College, the doctorate being taken under the spectroscopist who had taken the UCL appointment, and the war having emptied the observatory of the men who would otherwise have filled the benches. The caretaking appointment at the University of London Observatory being the exhibit: the young researcher having kept the telescopes and the plate vaults through the bombing years, the wartime blackout having darkened the London sky to an observing quality the city would never see again, and the night of August nineteen forty-four having produced the work that made the name, the spectroscopic observations benefiting from the darkened skies that the war itself supplied. The irony being the volume's own: the city's worst years having been the astronomer's best weather.

The discovery continuing across the postwar decade: the nineteen forty-five application to the Carnegie Institution's fellowships having been refused because the appointment carried observing time at Mount Wilson, the mountain being closed to women, and the refusal being the first entry in a ledger the rest of the volume keeps. The lectureship returning to London, the marriage to Geoffrey Burbidge in nineteen forty-eight, the pair having met at University College the year before, and the partnership of the two careers beginning at that meeting and never pausing for the following seventy years.

The crowning discovery arriving in the middle fifties: the invitation to Cambridge in nineteen fifty-five, the collaboration with the physicist William Fowler and the Yorkshireman Fred Hoyle having formed around the Burbidges' measurements of stellar chemical abundances, and the work having culminated in the nineteen fifty-seven synthesis paper whose initials, B2FH, became the field's shorthand for the origin of the elements. The paper and its making being the science chapter's center, and its personal center being the discovery this chapter keeps: the woman whose abundance tables made the paper possible having been the same woman the mountain had refused a decade before.

The London schooling being the discovery's middle movement: the grammar school places having been won by examination, the mathematics having been the girl's strongest subject from the first, and the University College having been the destination because the family being of the class that took the examinations seriously and the college being the institution that took the clever daughters the older universities still declined. The twenties and thirties of the capital being the discovery's backdrop: the city of the mother's cleaning work and the father's laboratory being the same city, and the girl having walked between the two economies daily, the walking being the biography's own geography lesson.

The doctoral years deserve their own paragraph: the thesis having been completed under the war's remaining turbulence, the University College having been partly evacuated, the observatory having been kept alive by the small staff that stayed, and the young researcher having been among the keepers, the keeping being the doctorate's real education, the education having included the practical trades the peacetime astronomer never learned: the generator that stalled, the dome that jammed in the ice, the blackout curtain that had to hang right, and the trades being the discovery's own apprenticeship, the apprenticeship having been served on the institution itself rather than on a subject, the subject having arrived later and having arrived prepared.

The meeting with the physicist being the discovery's personal hinge: the young theoretical physicist having attended the astronomy lectures at University College, the astronomer having been the more certain of the two about the career, the courtship of the late forties having been conducted between the lecture hall and the observatory, and the marriage of nineteen forty-eight having produced the field's most durable partnership, the partnership having been the discovery the chapter keeps returning to, the return being the biographer's own observation that the individual career being partly the story of the right collaborator arriving at the right hour, the hour being postwar London and the arrival being the whole life's turn.

The discovery chapter adding the eclipse exhibit: the nineteen twenty-seven eclipse having crossed the north of England, the seven-year-old having witnessed the darkness at midday, and the spectacle having been reported by the woman herself as the origin of the vocation, the vocation having been kept from that summer to the last

interview, the keeping being the discovery's own signature, the signature being dated and the date being the childhood summer the whole career traced its address to, the address being the sky rather than the observatory and the sky being the first teacher, the teacher having been total for a few minutes in June and the minutes having been enough.

The discovery closing on the invitation the woman's own career extended: the excluded student having become the included master, the master having kept the door open behind her, and the closing being the discovery's own moral, the moral being offered without sermon, the sermon being the one thing the shelf forbids itself, the forbidding being the discipline that keeps these biographies honest and the honesty being the reader's own protection and the protection being the last exhibit, the exhibit being the life itself and the life being open at the next page, the page being the beginning of the understanding and the understanding being the next chapter's assignment, the assignment being taken up in order and the order being the reader's own.

The discovery adding the mentor's line: the doctorate having been supervised by the era's spectroscopists at University College, the craft having been transmitted through the standard apprenticeships of the plate room, and the line being the discovery's genealogy, the genealogy being short because the craft being young, the young science having been handed from the founding generation to the Burbidge generation directly, the directly being the discovery's own luck and the luck being noted without embarrassment, the embarrassment being the modern reader's assumption and the assumption being wrong: the eras of small fields having been the eras of direct transmission, the transmission being the discovery's own quiet pedigree and the pedigree being honored here in the record's plain style.